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```
clc;
clear;
close all;
rng(1);
```

---------- INPUT PARAMETERS ----------

```
P.N = 64; % number of subcarriers
P.pilots = [8 22 44 58]; % pilot subcarrier indices (1-based)
P.Lcp = 16; % cyclic prefix length
P.Xp = [1+1j; 1-1j; -1+1j; -1-1j]/sqrt(2);
P.dataIdx = setdiff(1:P.N, P.pilots);

SNR_dB = 20;
NsymMain = 3000;
NsymSweep = 4000;
snrSweep = 0:5:30;

% 4-tap power-delay profile
pdp = exp(-(0:3)/1.5);
pdp = pdp/sum(pdp);
P.pdp = pdp(:);

% Channel frequency correlation matrix
dk = (0:P.N-1).' - (0:P.N-1);
R = zeros(P.N);

for l = 0:numel(pdp)-1
    R = R + pdp(l+1)*exp(-1j*2*pi*dk*l/P.N);
end

P.R = R;
```

---------- 1. MAIN RUN AT 20 dB ----------

```
out = run_ofdm(SNR_dB, NsymMain, P);
```

---------- 2. BER / MSE SWEEP ----------

```
berLS = zeros(size(snrSweep));
berMMSE = berLS;
berPerf = berLS;

mseLS = berLS;
mseMMSE = berLS;

for n = 1:numel(snrSweep)

    o = run_ofdm(snrSweep(n), NsymSweep, P);

    berLS(n) = o.ber(1);
    berMMSE(n) = o.ber(2);
    berPerf(n) = o.ber(3);

    mseLS(n) = o.mse(1);
    mseMMSE(n) = o.mse(2);

end

gam = 10.^(snrSweep/10)/2;

berTh = 0.5*(1 - sqrt(gam./(1+gam)));
```

---------- 3. PLOTS ----------

```
k = 1:P.N;

% Figure 1: Channel estimates

figure('Name','Channel estimates','Color','w', ...
    'Position',[40 40 1100 450]);

subplot(1,2,1);

plot(k, abs(out.H1(:,1)),'k','LineWidth',1.8);
hold on;

plot(k, abs(out.H1(:,2)),'r--','LineWidth',1.3);
```

```
plot(k, abs(out.H1(:,3)),'b-.','LineWidth',1.3);

plot(P.pilots, abs(out.Hpil1),'go', ...
    'MarkerFaceColor','g','MarkerSize',7);

grid on;
xlabel('Subcarrier index');
ylabel('|H(k)|');

title(sprintf('Channel magnitude (SNR = %d dB)', SNR_dB));

legend('Actual','LS (interpolated)','MMSE', ...
    'LS at pilots','Location','best');

subplot(1,2,2);

plot(k, angle(out.H1(:,1)),'k','LineWidth',1.8);
hold on;

plot(k, angle(out.H1(:,2)),'r--','LineWidth',1.3);
plot(k, angle(out.H1(:,3)),'b-.','LineWidth',1.3);

grid on;
xlabel('Subcarrier index');
ylabel('Phase (rad)');

title('Channel phase');

legend('Actual','LS','MMSE','Location','best');

% Figure 2: Estimation error

figure('Name','Estimation error','Color','w', ...
    'Position',[80 80 1100 420]);

subplot(1,2,1);

semilogy(k, out.mse_k(:,1),'r-o','MarkerSize',4);
hold on;

semilogy(k, out.mse_k(:,2),'b-s','MarkerSize',4);

xline(P.pilots,':k');

grid on;
xlabel('Subcarrier index');
ylabel('MSE = E|H - H_{est}|^2');

title(sprintf('Per-subcarrier estimation error (SNR = %d dB)', SNR_dB));

legend('LS','MMSE','Location','best');

subplot(1,2,2);
```

```
semilogy(snrSweep, mseLS,'r-o','LineWidth',1.4);
hold on;

semilogy(snrSweep, mseMMSE,'b-s','LineWidth',1.4);

grid on;
xlabel('SNR (dB)');
ylabel('Channel estimation MSE');

title('Estimation MSE vs SNR');

legend('LS','MMSE','Location','southwest');

% Figure 3: Equalised constellations

figure('Name','Constellations','Color','w', ...
    'Position',[120 120 1200 400]);

tit = {'Perfect CSI','LS estimate','MMSE estimate'};
fn = {'perfect','ls','mmse'};

for i = 1:3

    subplot(1,3,i);

    z = out.const.(fn{i});

    plot(real(z), imag(z), '.', 'MarkerSize', 3);
    hold on;

    ideal = [1+1j 1-1j -1+1j -1-1j]/sqrt(2);

    plot(real(ideal), imag(ideal),'r+', ...
        'MarkerSize',12,'LineWidth',2);

    grid on;
    axis equal;
    axis([-2 2 -2 2]);

    xlabel('In-phase');
    ylabel('Quadrature');

    berIdx = [3 1 2];

    title(sprintf('%s (BER = %.2e)', ...
        tit{i}, out.ber(berIdx(i))));

end

% Figure 4: BER vs SNR

figure('Name','BER comparison','Color','w', ...
    'Position',[160 160 700 500]);
```

```
semilogy(snrSweep, berLS,'r-o','LineWidth',1.5);
hold on;

semilogy(snrSweep, berMMSE,'b-s','LineWidth',1.5);
semilogy(snrSweep, berPerf,'k-^','LineWidth',1.5);
semilogy(snrSweep, berTh,'g--','LineWidth',1.2);

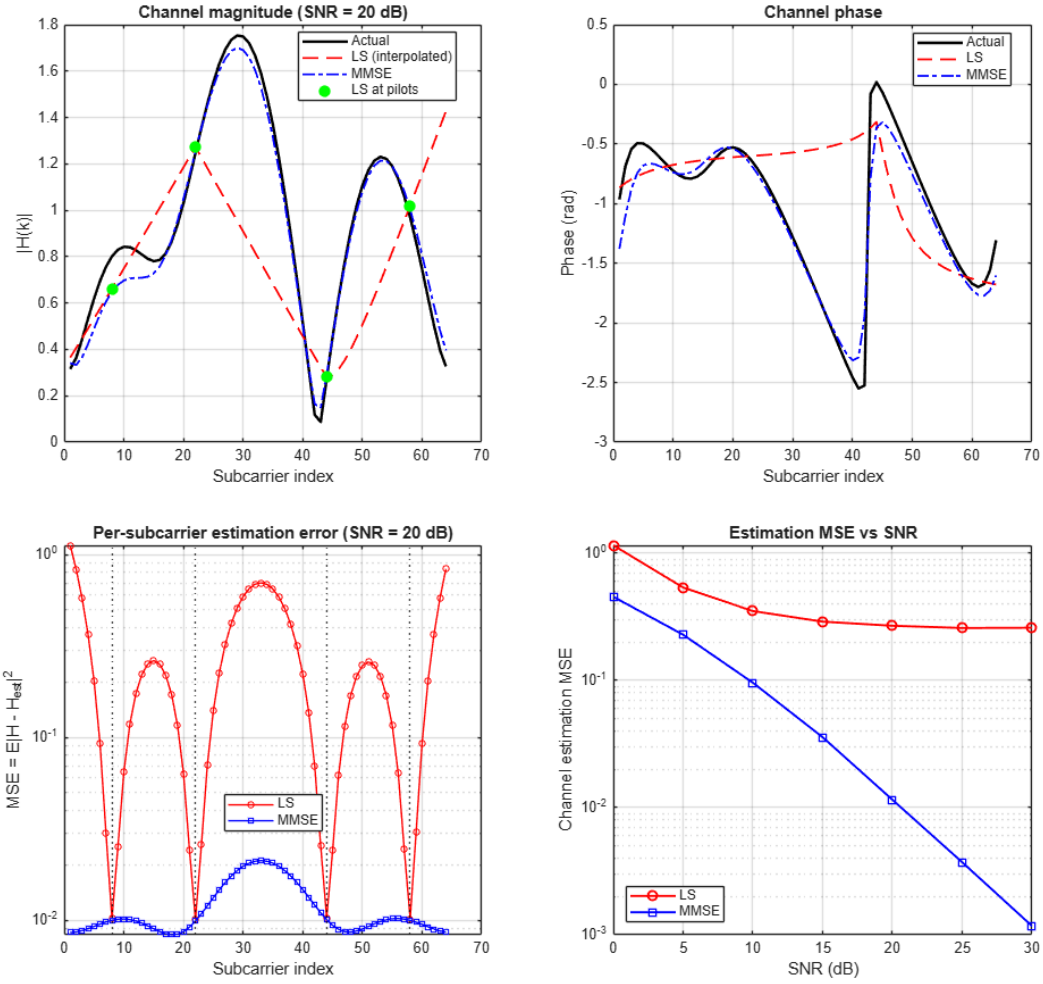
grid on;

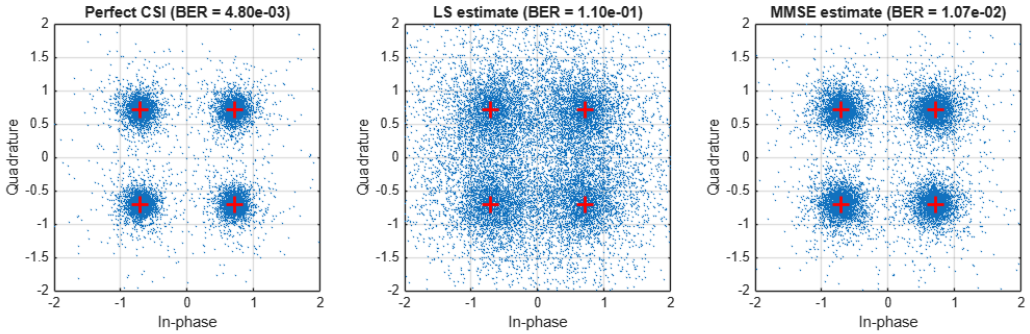
xlabel('SNR (dB)');
ylabel('Bit error rate');

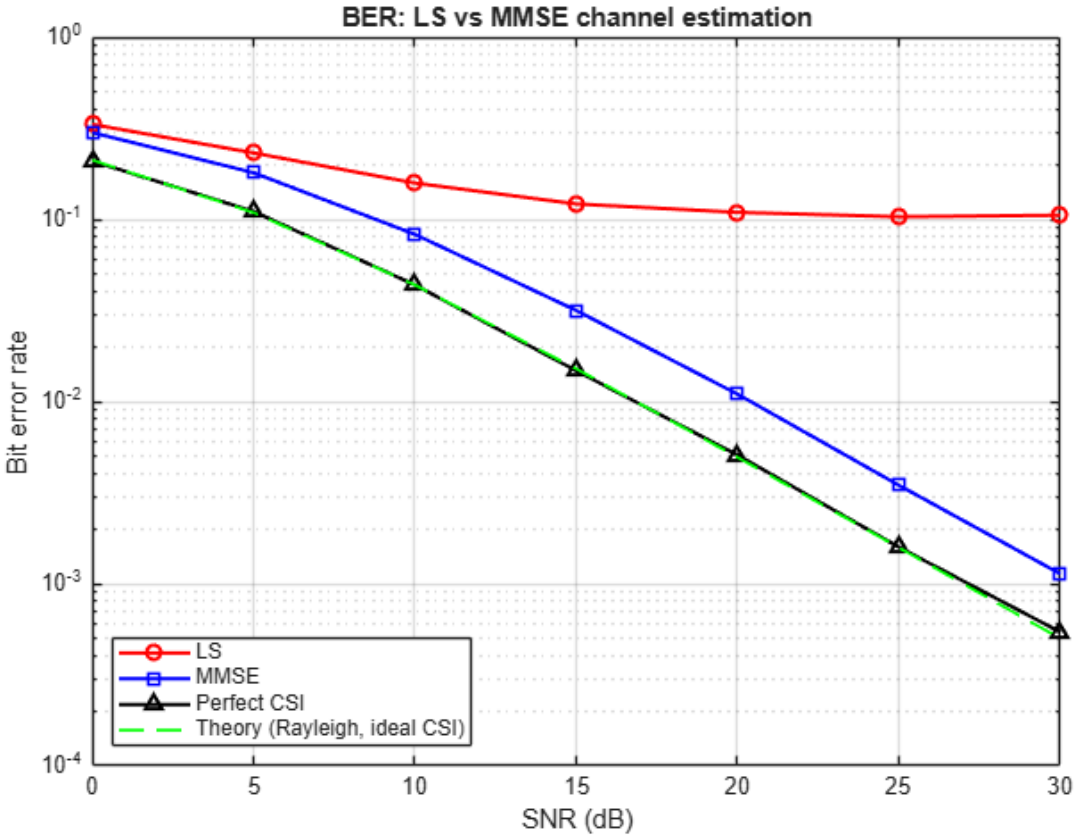
title('BER: LS vs MMSE channel estimation');

ylim([1e-4 1]);

legend('LS','MMSE','Perfect CSI', ...
    'Theory (Rayleigh, ideal CSI)','Location','southwest');
```







---------- 4. OUTPUT PARAMETER TABLE ----------

```
fprintf('\n========== OUTPUT PARAMETER TABLE (SNR = %d dB) ==========\n',
SNR_dB);

fprintf('\nChannel at pilot subcarriers (first OFDM symbol)\n');

fprintf('%-8s %-24s %-24s %-24s\n', ...
```

```
    'Pilot','Actual H','LS estimate','MMSE estimate');

for i = 1:numel(P.pilots)

    p = P.pilots(i);

    fprintf('%-8d %-24s %-24s %-24s\n', ...
        p, cstr(out.H1(p,1)), ...
        cstr(out.H1(p,2)), cstr(out.H1(p,3)));

end

fprintf('\n%-30s %-14s %-14s\n','Metric','LS','MMSE');

fprintf('%-30s %-14.4e %-14.4e\n', ...
    'Channel estimation MSE', out.mse(1), out.mse(2));

fprintf('%-30s %-14.4e %-14.4e\n', ...
    'Bit error rate (BER)', out.ber(1), out.ber(2));

fprintf('Perfect-CSI BER : %.4e\n', out.ber(3));

fprintf('Bits simulated at %d dB : %d\n', ...
    SNR_dB, 2*numel(P.dataIdx)*NsymMain);

fprintf('\n%-10s %-12s %-12s %-12s %-12s %-12s\n', ...
    'SNR (dB)','BER LS','BER MMSE','BER perfect','MSE LS','MSE MMSE');

for n = 1:numel(snrSweep)

    fprintf('%-10d %-12.3e %-12.3e %-12.3e %-12.3e %-12.3e\n', ...
        snrSweep(n), berLS(n), berMMSE(n), ...
        berPerf(n), mseLS(n), mseMMSE(n));

end


========== OUTPUT PARAMETER TABLE (SNR = 20 dB) ==========

Channel at pilot subcarriers (first OFDM symbol)
Pilot    Actual H                 LS estimate              MMSE
estimate
```

---------- LOCAL FUNCTIONS ----------

```
function s = cstr(z)

    s = sprintf('%.3f%+.3fj', real(z), imag(z));

end
```

```
function out = run_ofdm(snrdB, Nsym, P)

    N = P.N;
    pil = P.pilots(:);
    Np = numel(pil);

    dIdx = P.dataIdx(:);
    Nd = numel(dIdx);

    L = numel(P.pdp);

    sig2 = 10^(-snrdB/10);

    % Transmitter: QPSK data + pilots

    b1 = randi([0 1], Nd, Nsym);
    b2 = randi([0 1], Nd, Nsym);

    dataSym = ((1-2*b1) + 1j*(1-2*b2))/sqrt(2);

    X = zeros(N, Nsym);

    X(dIdx,:) = dataSym;
    X(pil,:) = repmat(P.Xp(:), 1, Nsym);

    x = sqrt(N)*ifft(X);

    xcp = [x(N-P.Lcp+1:N,:); x];

    % 4-tap Rayleigh channel + AWGN

    h = sqrt(P.pdp/2) .* ...
        (randn(L,Nsym) + 1j*randn(L,Nsym));

    y = zeros(N+P.Lcp, Nsym);

    for s = 1:Nsym

        y(:,s) = filter(h(:,s), 1, xcp(:,s));

    end

    y = y + sqrt(sig2/2)* ...
        (randn(size(y)) + 1j*randn(size(y)));

    % Receiver: remove CP, FFT

    Y = fft(y(P.Lcp+1:end,:))/sqrt(N);

    Hact = fft(h, N, 1);

    % LS estimate at pilots

    Hls_p = Y(pil,:) ./ P.Xp(:);
```

```
Hls = interp1(pil, Hls_p, (1:N).', ...
    'linear', 'extrap');

% MMSE channel estimation

Rpp = P.R(pil,pil);
Rfp = P.R(:,pil);

W = Rfp / (Rpp + sig2*eye(Np));

Hmmse = W * Hls_p;

% Equalisation

Yd = Y(dIdx,:);

Xe.perfect = Yd ./ Hact(dIdx,:);
Xe.ls = Yd ./ Hls(dIdx,:);
Xe.mmse = Yd ./ Hmmse(dIdx,:);

% BER calculation

nbits = 2*Nd*Nsym;

f = fieldnames(Xe);

ber = zeros(1,3);

for i = 1:3

    z = Xe.(f{i});

    err = sum(sum((real(z)<0) ~= (b1==1))) + ...
          sum(sum((imag(z)<0) ~= (b2==1)));

    ber(i) = err/nbits;

end

out.ber = [ber(2) ber(3) ber(1)];

% Channel estimation MSE

eL = abs(Hact - Hls).^2;
eM = abs(Hact - Hmmse).^2;

out.mse = [mean(eL(:)) mean(eM(:))];

out.mse_k = [mean(eL,2) mean(eM,2)];

% Items for plotting

out.H1 = [Hact(:,1) Hls(:,1) Hmmse(:,1)];
```

```
    out.Hpil1 = Hls_p(:,1);

    nPlot = min(300, Nsym);

    out.const.perfect = reshape( ...
        Xe.perfect(:,1:nPlot), [], 1);

    out.const.ls = reshape( ...
        Xe.ls(:,1:nPlot), [], 1);

    out.const.mmse = reshape( ...
        Xe.mmse(:,1:nPlot), [], 1);

end
```

```
8         0.649-0.469j             0.507-0.426j
0.499-0.414j
22        1.043-0.682j             1.048-0.722j
1.027-0.721j
44        0.274+0.005j             0.267-0.087j
0.274-0.103j
58        0.034-0.968j             -0.027-1.020j
-0.012-1.003j

Metric                          LS              MMSE
Channel estimation MSE          2.6876e-01      1.1712e-02
Bit error rate (BER)            1.0962e-01      1.0661e-02
Perfect-CSI BER : 4.8000e-03
Bits simulated at 20 dB : 360000

SNR (dB)    BER LS        BER MMSE      BER perfect   MSE LS        MSE MMSE
0           3.317e-01     2.993e-01     2.088e-01     1.152e+00     4.541e-01
5           2.313e-01     1.800e-01     1.103e-01     5.371e-01     2.285e-01
10          1.580e-01     8.261e-02     4.358e-02     3.506e-01     9.580e-02
15          1.213e-01     3.156e-02     1.473e-02     2.894e-01     3.571e-02
20          1.091e-01     1.095e-02     5.073e-03     2.687e-01     1.151e-02
25          1.032e-01     3.462e-03     1.583e-03     2.580e-01     3.714e-03
30          1.052e-01     1.131e-03     5.417e-04     2.592e-01     1.172e-03
```

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